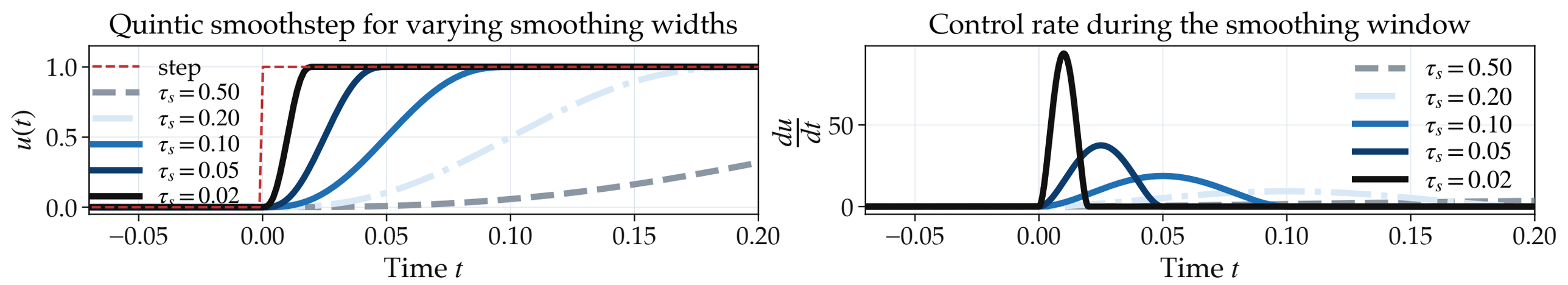
Quintic smoothstep for varying smoothing widths
u(t)
1.0
0.5
0.0
step
τs = 0.50
τs = 0.20
τs = 0.10
τs = 0.05
τs = 0.02
−0.05
0.00
0.05
0.10
0.15
0.20
Time t
Control rate during the smoothing window
du/dt
50
0
τs = 0.50
τs = 0.20
τs = 0.10
τs = 0.05
τs = 0.02
−0.05
0.00
0.05
0.10
0.15
0.20
Time t